25900147 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCNBC100 - BIOCHEMISTRY OF ENERGY TRANSFORMATION

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Symport carriers are also known as…..... [K] / [CO1]

   a). Co-enzymes b). Co-transporters

   c). Exchangers d). Transporters

2. List out the complexes in which iron sulphur atoms are present. [K] / [CO2]

   a). Complex I, II, III b). Complex IV, III

   c). Complex I, III d). Complex I ,III, IV

3. The process in which RNA is synthesised from DNA is called……. [K] / [CO3]

   a). Transcription b). Translation

   c). Replication d). Recombination.

4. Which enzyme model describes the active site as rigid and pre-shaped? [U] / [CO3]

a). Induced fit model

b). Koshlands theory

c). Lock and key model

d). substrate strain theory

5. Oxidative level phosphorylation takes place in [K] / [CO2]

a). cytoplasm b). inner mitochondrial membrane

c). cytoplasm d). mitochondrial matrix

6. Enzymes are ……..that increase the velocity of a reaction [K] / [CO3]

a). Biocatalyst b). biomolecules

c). bio active compounds d). high energy compounds

[1x6 = 6]

Part B

Short Answer Questions

Answer any **2** questions

Each question carries **2** marks

7. Explain Primary Active Transport? [U] / [CO1]

8. Describe ATP-ADP cycle [U] / [CO2]

9. Define Redox Potential? [K] / [CO2]

10. Define mRNA [K] / [CO3]

[2x2 = 4]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

11. Explain osmosis? [U] / [CO1]

12. Dicuss on Oxidation, Reduction and Redox Potential? [U] / [CO2]

13. Explain gene mapping and its classification [K] / [CO3]

14. Discuss on passive transport? [U] / [CO1]

15. Explain the stages of oxidation in food? [U] / [CO2]

16. Explain the different types of DNA repair [K] / [CO3]

[5x4 = 20]

Part D

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Recall the types of endocytosis? [K] / [CO1]

18. Explain high energy compounds and its classification? [U] / [CO2]

19. Explain Translation and post translational modifications [U] / [CO3]

20. Describe Recombinant DNA technology [K] / [CO3]

[10x2 = 20]